

Course Code	STA 7 CJ 401 (P)				
Course Title	ADVANCED ANALYTICAL TOOLS				
Type of Course	Major				
Semester	VII				
Academic Level	400-499				
Course Details	Credit	Lecture per week	Tutorial per week	Practicum per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic knowledge of Real analysis and Matrix theory.				
Course Summary	The main objective of this course to understand Reimann-Stieltjes integral, Uniform convergence, vector space Eigen values and Eigen vectors.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Evaluate the Riemann-Stieltjes integral and verify the conditions for the existence of the integrals.	An	C	Instructor-created exams
CO2	Demonstrate an understanding of limits and continuity of various functions	U	F	Practical Assignment
CO3	Develop skills in generalizing the concepts in univariate calculus to multivariate setup	Ap	C	Seminar Presentation
CO4	Demonstrate proficiency in understanding and applying vector space concepts	U	F	Instructor-created exams / Home Assignments
CO5	Demonstrate proficiency in applying vector space and matrix concepts	U	F	Writing assignments
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Riemann – Stieltjes Integral		10	15
	1	Definition, Linear properties- Integration by parts - Change of variable		
	2	Reduction to a Riemann integral		
	3	Step functions as integrators-Reduction to a finite sum		
	4	Monotonically increasing integrators- Riemanns conditions-Comparison theorems- Functions of bounded variations (concepts only)		
	5	Necessary & Sufficient conditions for the existence of Riemann Stieltjes integral		
	6	Mean-value theorems		
II	Sequences and Series of Functions		13	20
	7	Point wise convergence of sequence of functions - Examples of sequences of real valued functions		
	8	Definition of Uniform convergence - Uniform convergence and continuity		
	9	Cauchy condition for uniform convergence - Uniform convergence of infinite series of functions		
	10	Uniform convergence and Riemann-Stieltjes Integration - Uniform convergence and differentiation		
	11	Multivariable Functions- Limits and continuity of multivariable functions – Derivatives - directional derivatives		
	12	Total derivative in terms of partial derivatives		
III	Algebra of Vectors		10	15
	14	Vector spaces - definition and examples		
	15	Subspaces - Linear independence - Basis and dimension- Linear equations		
	16	Vector spaces with an inner product: Properties		
	17	Gram-Schmidt orthogonalization.		
IV	Algebra of matrices		12	20
	18	Theory of matrices and determinants - Matrix operations-Elementary matrices and diagonal reduction of a matrix- Determinants-Transformations		
	19	Generalized inverse of a matrix		
	20	Matrix representations of vector spaces, bases, etc.		
	21	Idempotent matrices. Special products of matrices		
	22	Eigene values and reduction of matrices: Classification and transformations of quadratic forms. Roots of determinant equations. Canonical reduction of matrices.		
V	Open Ended Module:		30	
		Hands-on-activities using Python/R. Open book problem solving exercises		

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Text Books

1. **Khuri, A.T. (1993).** Advanced Calculus with Applications in Statistics. John Wiley & Sons, New York.
(Chapter 7).
2. **Apostol, T.M. (1974).** Mathematical Analysis- Second Edition. Narosa Publications, New Delhi.
3. **Rao, C.R. (2002).** Linear Statistical Inference & Its Applications- Second Edition. John Wiley & Sons, New York.
4. **Rao, A.R. & Bhimasankaram, P. (1992).** Linear Algebra. Hindustan Book Agency, New Delhi.
5. **Rao, A.R. and Bhimsankaram, P. (1992).** Linear Algebra. Tata McGraw Hill, New Delhi.
6. **Lewis, D.W. (1996).** Matrix Theory. Allied Publishers, Bangalore.
7. **Graybill, F. A. (1983).** Matrices with Applications in Statistics. John Wiley & Sons, New York.

References:

1. **Widder, D.A. (1996).** Advanced Calculus, Second Edition, Prentice Hall, Inc., New Delhi.
2. **Malik, S.C. & Arora, S. (2006).** Mathematical Analysis- Second Edition. New Age International, New Delhi.
3. **Rudin, W. (1976).** Principles of Mathematical Analysis- Third Edition. McGraw Hill, New York
4. **Biswas, S. (1997).** A text book of Linear Algebra. New Age International, New Delhi.
5. **Rao, C.R. (2002).** Linear Statistical Inference and Its Applications- Second Edition. John Wiley & Sons, New York.